Duy D. VU

Full-stack Protein Designer

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WORKING EXPERIENCE

1/2025 – present: Full-stack Protein Designer

Computational Structural Biology Unit, Institut Pasteur, Paris, France

- Leading end-to-end design, production, and experimental validation of 700+ protein constructs, achieving multiple successful design campaigns.
- Implementing state-of-the-art protein design pipelines across diverse applications:
  – Targeted degradation of TDP-43 for ALS (collab. with Zoher Gueroui, École Normale Supérieure)
  – Allosteric control of fibril condensates (collab. with Keren Lasker, Scripps Research)
  – Functionalization of AAV vectors for targeted delivery (collab. with Breden Evano, Institut Pasteur)
  – Stabilization of megacomplexes for cryo-EM (collab. with Filippo Mancia, Columbia University)
  – Designing nanobodies/minibinders against Hantavirus (collab. with Timothée Bruel and Pablo Guardado Calvo, Institut Pasteur)
- Developing a novel design pipeline for dual-target protein binder for interacting motifs.
- Developing a novel pipeline for designing fold-switching protein
- Building an integrated experimental platform (cell-free expression + BLI) to accelerate screening and validation of designed proteins.
- Performing large-scale structural predictions and data analysis, creating interactive dashboards to generate testable biological hypotheses.
- Developing a geometric deep learning model for classification and clustering of interacting motifs

8/2026: Visiting Researcher in Protein Design

DeGrado Lab, UCSF, San Francisco, USA

- Developing a novel pipeline for designing radioactive protein binder cage for targeted delivery in cancer radiotherapy.

09/2019 – 12/2024: Structural Biologist Focusing on Disordered Proteins

Laboratoire des biomolécules, UMR 7203, Paris, France

- Revealed the molecular interactions underlying the liquid-liquid phase separation of DNA double-strand break repair by NHEJ machinery.
- Elucidated the binding mechanism of an intrinsically disordered protein using chemical exchange NMR.
- Characterized the nanosecond dynamics of intrinsically disordered proteins using high-resolution NMR relaxometry.
- Established and optimized protocols for labeling fluorine in proteins for NMR studies.

EDUCATION

PhD in Integrative Structural Biology *Ecole Normale Superieur – PSL university, Paris, France*

- Supervising 3 master students and 2 bachelor students

Master in Integrative Structural Biology *Université Grenoble Alpes, Grenoble, Franc*

Bachelor in Pharmacy *University of Medicine and Pharmacy, Ho Chi Minh city, Viet Nam*

RESEARCH SKILLS

- **Protein Design**: Protein structure modeling, high-throughput protein-protein interaction prediction, binder design, and sequence design to improve stability/solubility by deep learning

with both hallucination and diffusion approaches (AlphaFold, Boltz, RFdiffusion, BindCarft, proteinMPNN, Pyrosetta, Boltzgen, PXdesign, Protena-Complexa).

- **Biophysics:** Isotope labelling of proteins (uniform or selective) for structural characterization, NMR assignment, and characterizing the kinetics and dynamics of protein/DNA-protein interactions using various approaches (NMR titration, 2D lineshape analysis, PRE, CPMG, CEST), NMR relaxometry, ITC, FP, MTS, DSC, BLI and SPR
- **Biochemistry**: Extensive experience in protein expression and purification in bacteria, including the optimization and trouble-shooting of protocols; proficient in using the AKTA system for affinity, ion-exchange, and gel filtration chromatography; protein analysis using SDS-PAGE, Western blot, and SEC-MALLS; experienced in fluorescent protein labeling and utilizing epifluorescence microscopy to monitor liquid-liquid phase separation (LLPS), protein qualitity control (MS, DSL, UV-VIS).

Languages

- Vietnamese: mother tongue
- English: full professional proficiency
- French: sufficient working proficiency

SCIENTIFIC PUBLICATION

Publications:

- "*Molecular characterization of MRI/CYREN reveals the Ku binding mode and the role of multimerization in stimulating the activity of NHEJ in DNA repair.*" **Duc-Duy VU** et al. **bioRxiv**, pp.2026-07.
- "*Ku70-SAP domain has an overlapping function with DNA-PKcs in limiting the lateral movement of the Ku ring along DNA*." Zhu,…**Duc-Duy VU**,…. **Nucleic Acids Research**, *54*(13), p.gkag670.
- "*BPS2026–De novo design of protein binders for programmable control of biomolecular condensates.*" **Duc-Duy VU**, *et al.* ***Biophysical Journal*** 125.4 (2026): 114a.
- "*Multivalent interactions of the disordered regions of XLF and XRCC4 foster robust cellular NHEJ and drive the formation of ligation-boosting condensates in vitro*" **Duc-Duy VU** *et al.*, **Nature Structural & Molecular Biology 31.11 (2024): 1732-1744.**
- "*Structure and dynamics of the unassembled nucleoprotein of rabies virus in complex with its phosphoprotein chaperone module*" Francine …., **Duc-Duy VU,…** *et al.* **Viruses** 14, no. 12 (2022): 2813.

Scientific peer review: Nature Communications, Nature Biotechnology

PhD thesis: *Functions and Dynamics of Intrinsically Disordered Protein Regions in Non-homologous End-Joining by NMR spectroscopy*

REFERENCES

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